Structure-Constrained Behavior Generation: A Minimal Evidence for Structure-Dominant Behavioral Dynamics

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Abstract

We propose a minimal framework for behavior generation in dynamical systems, termed Structure-Constrained Behavior Generation (SCBG). Unlike traditional control paradigms that derive behavior from optimization or learned policies, SCBG posits that behavior emerges as a selection from a structure-constrained set of reachable modes.

We construct a minimal 2D agent model with fixed local update rules and compare three structural configurations: a structured “closure” topology, a dense topology, and random connectivity. Across controlled stimulus conditions, we observe that behavioral outcomes, success rates, and stability metrics vary significantly with structural configuration, even under identical update rules.

Further perturbation experiments demonstrate that altering key structural relations leads to substantial degradation of behavioral performance, while perturbations to input mappings have comparatively weaker effects. This suggests that structure acts as a primary determinant of behavioral generation, with input acting as a modulation factor.

We present this as a minimal evidence chain supporting the hypothesis that behavior can be generated through structural constraints rather than direct policy optimization.

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1. Introduction

Modern control systems typically derive behavior through one of the following paradigms: • Analytical control laws (e.g., PID, MPC) • Optimization-based planning • Data-driven policy learning (e.g., reinforcement learning)

In all these approaches, behavior is primarily determined by:

$$\text{State} \rightarrow \text{Policy / Optimization} \rightarrow \text{Action}$$

However, recent developments in biological modeling and network-based systems suggest that structure itself may play a more fundamental role in shaping behavior.

This work explores an alternative formulation:

Behavior is not directly computed or learned, but emerges from constraints imposed by system structure.

We term this framework Structure-Constrained Behavior Generation (SCBG).

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1. Core Hypothesis

We define the minimal SCBG formulation as:

$$S + I + U \Rightarrow B$$

Where: • S: Structural configuration • I: External input (stimulus) • U: Local update rule • B: Behavioral distribution

The key hypothesis is:

Under fixed update rules, variations in behavior are primarily driven by structural differences rather than input mappings or optimization processes.

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1. Minimal Model

3.1 Environment

We consider a 2D agent operating in a simple environment consisting of: • A target location • A single obstacle • Boundary constraints

3.2 Node Structure

The agent is governed by a minimal 4-node system: • A: Approach node • B: Avoidance node • C: Stabilization node • D: Switching node

Each node maintains a scalar activation value updated over time.

3.3 Update Rule

Node activations evolve according to:

z_i(t+1) = \tanh\left(\sum_j W_{ij} z_j + \sum_k G_{ik} I_k - \lambda_i z_i\right)

Where: • W: Structural connectivity • G: Input mapping • I: Stimulus vector • \lambda: leakage

3.4 Action Mapping

Agent motion is derived from node activations: • Forward velocity depends on approach and stabilization • Angular velocity depends on avoidance and switching

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1. Experiments

4.1 Structural Comparison (v1)

We compare three structures: 1. Closure (structured topology) 2. Dense (fully connected baseline) 3. Random (random connectivity)

Under identical update rules and stimuli.

4.2 Structural Perturbation (v2)

We perturb the closure structure via: • Edge pruning • Sign flipping • Noise injection • Sparsification

4.3 Structure vs Input Mapping (v3)

We compare: • Perturbations in structural matrix W • Perturbations in input mapping G

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1. Results

5.1 Structural Dependence

We observe that: • Closure structure achieves consistent success across all stimuli • Dense structure performs inconsistently • Random structure fails across all conditions

5.2 Structural Sensitivity

Perturbations to closure structure result in: • Complete failure under sign inversion • Significant degradation under noise • Reduced generalization under sparsification

5.3 Structure vs Input Dominance

Comparative analysis shows:

$$\Delta(W) \gg \Delta(G)$$

Where: • Structural perturbations cause large behavioral degradation • Input mapping perturbations cause limited modulation

This supports the claim that:

Structure acts as the primary determinant of behavior generation.

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1. Discussion

The SCBG framework suggests a shift in perspective: • Behavior is not computed from scratch • Behavior is selected from a structure-constrained space

This differs fundamentally from: • Optimization-based control • Policy learning

Instead, SCBG aligns more closely with: • Mode selection • Dynamical attractor systems • Constraint-based generation

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1. Limitations

This work is limited to: • A minimal 2D agent • Simplified node dynamics • No real-world robotic validation • No comparison with advanced RL baselines

Therefore, this should be viewed as:

A minimal evidence study, not a complete control theory.

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1. Conclusion

We present initial evidence that:

Behavior can emerge as a structure-constrained selection process rather than direct optimization or learning.

This suggests a potential new direction for control systems and behavior generation, where structural design plays a central role.
